

AP Precalculus Unit 2 - Exponential and Logarithmic Functions

Complete Guided Notes

15 strengthened topic packets

Purpose

This volume combines the fully rewritten guided notes for Topics 2.1-2.15. Each packet contains learning targets, vocabulary, a reasoning launch, three concept modules, worked examples, independent checks, and an exit ticket.

Resource design

- Original, non-repeating questions with a deliberate progression from foundational structure to modeling and justification.
- Explicit calculator guidance, exact-value expectations, domain restrictions, parameter interpretation, and error analysis.
- AP-style multiple-choice and free-response tasks integrated into every topic.
- Consistent ECHS formatting for classroom printing, annotation, and digital distribution.

Topics included

Topic	Title
2.1	Change in Arithmetic and Geometric Sequences
2.2	Change in Linear and Exponential Functions
2.3	Exponential Functions
2.4	Exponential Function Manipulation
2.5	Exponential Function Context and Data Modeling
2.6	Competing Function Model Validation
2.7	Composition of Functions
2.8	Inverse Functions
2.9	Logarithmic Expressions
2.10	Inverses of Exponential Functions
2.11	Logarithmic Functions
2.12	Logarithmic Function Manipulation
2.13	Exponential and Logarithmic Equations and Inequalities
2.14	Logarithmic Function Context and Data Modeling
2.15	Semi-log Plots



ECHS Mathematics
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Topic 2.1 - Change in Arithmetic and Geometric Sequences

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Why this topic matters

Sequences are functions whose inputs are usually whole-number indices. The strongest solutions identify the invariant that drives the sequence, state the domain, and coordinate at least two representations rather than relying on visual pattern recognition alone.

Learning targets

- Describe arithmetic change as repeated addition and geometric change as repeated multiplication.
- Move among recursive, explicit, numerical, graphical, and contextual representations.
- Use discrete domains, precise notation, and contextual units.

Language and structure

Term	Meaning in this topic
common difference	The constant amount added to obtain the next term of an arithmetic sequence.
common ratio	The constant factor used to obtain the next term of a geometric sequence.
recursive rule	A rule that defines a term from one or more earlier terms.
explicit rule	A rule that gives a term directly from its index.
discrete domain	An input set made of separated values, usually whole-number indices.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Two sequences begin at 12. Sequence A adds 7 each step. Sequence B multiplies by 1.35 each step. Without calculating more than four terms, predict which sequence will eventually be larger and explain what evidence would justify the claim.

Concept 1: Arithmetic change: equal additions

Core idea

If $a_{n+1} - a_n = d$ for every allowed index, then the sequence is arithmetic.

- Recursive form: $a_1 = A$, $a_n = a_{n-1} + d$ for $n \geq 2$.
- Explicit form: $a_n = A + d(n - 1)$.
- A positive d means the terms increase by a constant amount; it does not guarantee every term is positive.

Worked example - annotate the reasoning

Problem. An arithmetic sequence satisfies $a_4 = 23$ and $a_{11} = 58$. Determine d , a_1 , and an explicit rule.

Model reasoning. The index gap is $11 - 4 = 7$, so $58 - 23 = 7d$ and $d = 5$. Then $a_1 = 23 - 3(5) = 8$. Thus $a_n = 8 + 5(n - 1) = 5n + 3$.

Check your understanding

Independent

The sequence b_n has $b_3 = -2$ and $b_9 = 34$. Find b_1 and b_n .

Concept 2: Geometric change: equal multiplications

Core idea

If $\frac{g_{n+1}}{g_n} = r$ for every allowed index with $g_n \neq 0$, then the sequence is geometric.

- Recursive form: $g_1 = G$, $g_n = r g_{n-1}$ for $n \geq 2$.
- Explicit form: $g_n = G r^{n-1}$.
- For $0 < r < 1$ and $G > 0$, the sequence stays positive and decreases toward 0.

Worked example - annotate the reasoning

Problem. A positive geometric sequence has $g_2 = 12$ and $g_5 = 324$. Determine r , g_1 , and g_n .

Model reasoning. The index gap is 3, so $324/12 = r^3 = 27$ and $r = 3$. Then $g_1 = 12/3 = 4$, so $g_n = 4(3)^{n-1}$.

Check your understanding

Independent

A geometric sequence has $c_2 = -6$ and $c_5 = 48$. Find the real value of r and an explicit rule.

Concept 3: Representations and domain

Core idea

A finite list supports a model only after a sequence family or defining rule is stated.

- Sequence graphs use isolated points because intermediate indices are not automatically allowed.
- A table can reveal constant differences or ratios, but several formulas can agree on finitely many displayed terms.
- In context, the index, initial value, and units determine whether the first term is a_0 or a_1 .

Worked example - annotate the reasoning

Problem. A savings account begins with 500 QAR at month 0 and grows by 8% each month. Write recursive and explicit rules, then state the natural domain for the first year.

Model reasoning. $A_0 = 500$ and $A_n = 1.08A_{n-1}$. Explicitly, $A_n = 500(1.08)^n$. For the first year the natural domain is $n \in \{0, 1, \dots, 12\}$, measured in whole months.

Check your understanding

Independent

Explain why connecting the points of the sequence $s_n = 2^n$ can create a misleading graph.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Classify $18, 12, 8, \frac{16}{3}, \dots$ and write an explicit rule beginning at $n = 1$.

Exit ticket 2

No calculator

An arithmetic sequence has $u_7 = 41$ and common difference -4 . Find u_1 .

Exit ticket 3

No calculator

State one reason the terms $2, 5, 10, 17$ do not by themselves prove a unique continuation.

Topic 2.2 - Change in Linear and Exponential Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Why this topic matters

Linear and exponential functions can both increase or decrease, but they organize change differently. A linear model adds the same amount over equal input intervals; an exponential model multiplies by the same factor. Strong comparisons focus on the invariant and on what the parameters mean in context.

Learning targets

- Distinguish additive change from proportional change across tables, graphs, formulas, and contexts.
- Construct linear and exponential functions from two values or from stated rates.
- Compare models, interpret intersections, and respect discrete versus continuous domains.

Language and structure

Term	Meaning in this topic
additive change	Equal input intervals produce equal output differences.
proportional change	Equal input intervals produce equal output ratios or percent changes.
initial value	The function value at the chosen reference input, often $x = 0$.
growth factor	The multiplier b in ab^x ; the percent change is $(b - 1)100\%$.
intersection	An input-output pair satisfying both models; in context it marks equal predicted values.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A school campaign can gain supporters in two ways. Plan L starts with 80 supporters and adds 26 each week. Plan E starts with 80 supporters and grows by 22% each week. Which plan is larger during the first month, and which is likely larger later? State what must be calculated rather than guessed.

Concept 1: Recognizing the invariant

Core idea

Linear change has constant differences; exponential change has constant ratios for equal input intervals.

- Linear form: $L(x) = mx + b$, where m is change in output per one input unit.
- Exponential form: $E(x) = ab^x$, where b is the factor per one input unit.
- A graph's shape alone is not enough; use coordinates, scale, or a defining rule.

Worked example - annotate the reasoning

Problem. A table contains $(0, 18), (2, 30), (4, 42), (6, 54)$. Determine whether a linear or exponential model is appropriate and write the model.

Model reasoning. Every increase of 2 in x adds 12 to y , so the rate is 6 per input unit. The model is $y = 18 + 6x$.

Check your understanding

Independent

The values of y at $x = 0, 1, 2, 3$ are 5, 7.5, 11.25, 16.875. Identify the model family and write a rule.

Concept 2: Building models from two values

Core idea

For a linear model use slope; for an exponential model use a ratio adjusted for the input gap.

- Linear: $m = \frac{y_2 - y_1}{x_2 - x_1}$, then solve for b .
- Exponential: $\frac{y_2}{y_1} = b^{x_2 - x_1}$, then solve for a .
- Units belong to m ; the growth factor b is dimensionless but tied to a time interval.

Worked example - annotate the reasoning

Problem. Construct a linear function through $(3, 17)$ and $(9, 47)$, and an exponential function through $(1, 48)$ and $(4, 384)$.

Model reasoning. Linear: $m = (47 - 17)/(9 - 3) = 5$, so $L(x) = 5x + 2$. Exponential: $384/48 = 8 = b^3$, so $b = 2$ and $48 = a(2)$, giving $E(x) = 24(2)^x$.

Check your understanding

Independent

An exponential function satisfies $f(2) = 75$ and $f(6) = 1200$. Find $f(x)$.

Concept 3: Comparing models and interpreting intersections

Core idea

The model with the larger current value need not have the larger long-run value.

- Compare functions at the same inputs and identify the first input at which the ordering changes.
- An intersection means equal model predictions, not that the real quantities must be equal.
- Extrapolation is only defensible while the assumptions and context remain reasonable.

Worked example - annotate the reasoning

Problem. Plans A and B cost $A(t) = 120 + 35t$ and $B(t) = 120(1.18)^t$ QAR after t months. Find the first whole month when B costs more than A.

Model reasoning. At $t = 6$, $A = 330$ and $B \approx 323.95$. At $t = 7$, $A = 365$ and $B \approx 382.26$. Therefore the first whole month is 7.

Check your understanding

Independent

Explain why the statement “exponential always grows faster than linear” is incomplete.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1**No calculator**

Identify the model family and rule for the table $(0, 240)$, $(1, 204)$, $(2, 173.4)$, $(3, 147.39)$.

Exit ticket 2**No calculator**

A linear function passes through $(4, -7)$ and $(10, 20)$. Find its rate of change.

Exit ticket 3**No calculator**

Interpret the intersection of two revenue models in context.

Topic 2.3 - Exponential Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

An exponential function is controlled by a starting scale, a multiplicative base, and possible transformations. A complete analysis names the horizontal asymptote, specifies whether the graph approaches it from above or below, and distinguishes growth or decay from increasing or decreasing behavior after reflections.

Learning targets

- Analyze the structure, domain, range, monotonicity, concavity, and end behavior of exponential functions.
- Connect equal-interval proportional change to the parameters of an exponential formula.
- Describe and construct transformed exponential functions using precise asymptotic language.

Language and structure

Term	Meaning in this topic
base	The positive constant $b \neq 1$ in ab^x that controls proportional change.
horizontal asymptote	A horizontal line approached by the graph as x moves toward an end of the domain.
monotonicity	Whether a function is increasing or decreasing over an interval.
concavity	Whether slopes are increasing or decreasing; exponential graphs do not have turning points.
transformation	A shift, reflection, stretch, or compression applied to a parent function.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Two functions have the same horizontal asymptote $y = 6$. One passes through $(0, 9)$ and rises; the other passes through $(0, 3)$ and falls as x increases. Sketch a plausible shape for each and explain which parameter signs are forced.

Concept 1: The parent family $f(x) = ab^x$

Core idea

For $a \neq 0$, $b > 0$, and $b \neq 1$, the domain is all real numbers and the horizontal asymptote is $y = 0$.

- If $a > 0$ and $b > 1$, the function is positive, increasing, and concave up.
- If $a > 0$ and $0 < b < 1$, the function is positive, decreasing, and concave up.
- A negative a reflects the graph across the x -axis, reversing the sign and monotonicity.

Worked example - annotate the reasoning

Problem. Analyze $f(x) = 5(1/2)^x$: state domain, range, intercept, monotonicity, concavity, asymptote, and end behavior.

Model reasoning. Domain: all real numbers. Range: $(0, \infty)$. y -intercept $(0, 5)$. The function is decreasing and concave up. Horizontal asymptote: $y = 0$. As $x \rightarrow \infty$, $f(x) \rightarrow 0^+$; as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

Check your understanding

Independent

Analyze $g(x) = -3(4)^x$ without graphing.

Concept 2: Transformations and asymptotes

Core idea

In $f(x) = ab^{x-h} + k$, the horizontal asymptote is $y = k$ and the anchor point is $(h, a + k)$.

- h shifts the graph horizontally; k shifts it vertically.
- The sign of a determines whether values lie above or below the asymptote.
- The range is (k, ∞) when $a > 0$ and $(-\infty, k)$ when $a < 0$.

Worked example - annotate the reasoning

Problem. Describe the graph of $f(x) = -2(3)^{x-1} + 4$, including the asymptote, range, intercepts, and end behavior.

Model reasoning. Asymptote $y = 4$; range $(-\infty, 4)$. Anchor point $(1, 2)$. y -intercept is $-2/3 + 4 = 10/3$. The x -intercept satisfies $3^{x-1} = 2$, so $x = 1 + \log_3 2$. The graph decreases, approaches 4 from below as $x \rightarrow -\infty$, and tends to $-\infty$ as $x \rightarrow \infty$.

Check your understanding

Independent

For $p(x) = 7(0.4)^{x+2} - 5$, state the asymptote, anchor point, range, and monotonicity.

Concept 3: Constructing an exponential function from features

Core idea

When a horizontal asymptote is known, subtract it before using ratios.

- If $f(x) = ab^x + k$, then $f(x) - k = ab^x$.
- Two points with known k determine a and b when the adjusted outputs are nonzero and have a consistent ratio.
- Check that the resulting base is positive and not equal to 1.

Worked example - annotate the reasoning

Problem. An exponential function has horizontal asymptote $y = 2$ and passes through $(0, 8)$ and $(2, 56)$. Find the function.

Model reasoning. Use $f(x) = ab^x + 2$. From $f(0) = 8$, $a = 6$. From $f(2) = 56$, $6b^2 = 54$, so $b = 3$. Therefore $f(x) = 6(3)^x + 2$.

Check your understanding

Independent

Find $f(x) = ab^x - 4$ if $f(1) = 2$ and $f(3) = 50$.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

State the range and horizontal asymptote of $f(x) = -7(0.2)^{x-3} + 9$.

Exit ticket 2

No calculator

Find the anchor point of $g(x) = 4(5)^{x+1} - 2$.

Exit ticket 3

No calculator

Why can an exponential graph not cross its horizontal asymptote in the form $ab^{x-h} + k$ with $a \neq 0$?

Topic 2.4 - Exponential Function Manipulation

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

Manipulating exponential functions is not cosmetic: a useful form can reveal an initial value, a factor for a new time unit, or a graph transformation. Every rewrite should preserve the function and make one feature easier to interpret.

Learning targets

- Rewrite exponential expressions using exponent properties and common bases.
- Translate growth factors across equivalent input intervals using rational exponents.
- Explain how algebraic equivalence corresponds to shifts, stretches, and unchanged graphs.

Language and structure

Term	Meaning in this topic
equivalent form	A different expression that gives the same output for every input in the domain.
rational exponent	An exponent such as $1/12$ that represents a root and supports time-scale conversion.
common base	A shared base used to compare or combine exponential expressions.
input scaling	Replacing x by cx , which changes the factor associated with one unit of input.
horizontal shift	Replacing x by $x - h$; exponent laws can convert the shift into a vertical scale factor.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Without using a calculator, decide which of the following are the same function: $f(x) = 4^{x+1}$, $g(x) = 16 \cdot 4^{x-1}$, and $h(x) = 2^{2x+2}$. Justify by rewriting rather than checking a few values.

Concept 1: Exponent laws and common bases

Core idea

Use $b^{u+v} = b^u b^v$, $b^{u-v} = b^u / b^v$, and $(b^u)^v = b^{uv}$.

- Rewrite numbers such as 8, 16, and 32 as powers of 2 when a common base is helpful.
- Coefficients outside an exponential cannot generally be moved into the exponent unless they are themselves powers of the base.
- Check equivalence by simplifying both expressions to the same coefficient, base, and exponent.

Worked example - annotate the reasoning

Problem. Show that $12(4)^{x-2}$, $\frac{3}{4}(4)^x$, and $3(2)^{2x-2}$ are equivalent.

Model reasoning. $12(4)^{x-2} = 12(4^x/16) = \frac{3}{4}4^x$. Also $3(2)^{2x-2} = 3(2^{2x}/4) = \frac{3}{4}(4^x)$.

Check your understanding

Independent

Rewrite $5(8)^{x+1}$ in the form $A(2)^{3x}$.

Concept 2: Changing the time unit

Core idea

If a factor R applies over k input units, then the one-unit factor is $R^{1/k}$.

- Annual factor from monthly factor m : m^{12} .
- Monthly factor from annual factor A : $A^{1/12}$.
- Always attach the factor to its interval; 1.08 per year and 1.08 per month describe very different change.

Worked example - annotate the reasoning

Problem. A population grows by a factor of 1.60 every 5 years. Write a model using a one-year growth factor.

Model reasoning. If $P(t) = P_0 b^t$ with t in years, then $b^5 = 1.60$, so $b = 1.60^{1/5}$. Thus $P(t) = P_0(1.60^{1/5})^t = P_0(1.60)^{t/5}$.

Check your understanding

Independent

An account grows by 2% per month. Write the equivalent annual growth factor and annual percent increase.

Concept 3: Algebraic form and graph transformations

Core idea

A horizontal shift in an exponential can be rewritten as a vertical scaling because $b^{x-h} = b^{-h}b^x$.

- $ab^{x-h} = ab^{-h}b^x$; the graph is unchanged.
- Replacing x by cx changes the effective one-unit factor from b to b^c .
- Two expressions can describe the same graph while highlighting different parameters.

Worked example - annotate the reasoning

Problem. Rewrite $f(x) = 7(3)^{x-2} + 4$ in the form $A(3)^x + 4$ and explain the graph equivalence.

Model reasoning. $7(3)^{x-2} + 4 = \frac{7}{9}(3)^x + 4$. Both expressions produce the same graph; the first highlights a right shift of 2, while the second highlights a vertical scale coefficient $7/9$.

Check your understanding

Independent

Find c so that $2(5)^{3x} = 2(c)^x$ for all real x .

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1**No calculator**

Rewrite $9(27)^{x-1}$ using base 3 and simplify the coefficient.

Exit ticket 2**No calculator**

A factor of 1.44 applies every 2 years. Find the annual factor.

Exit ticket 3**No calculator**

Explain why $2^{x+3} \neq 2^x + 2^3$.

Topic 2.5 - Exponential Function Context and Data Modeling

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Why this topic matters

Exponential modeling begins with a multiplicative mechanism, not merely a curved graph. A strong model identifies the reference value, the factor and its interval, the input domain, and the assumptions that make repeated proportional change reasonable.

Learning targets

- Construct exponential models from percent rates, two points, transformations, or regression output.
- Use exponential models to predict outputs and solve for inputs.
- Interpret parameters, equivalent time scales, domains, units, and model limitations.

Language and structure

Term	Meaning in this topic
growth factor	The multiplier greater than 1 applied over a specified interval.
decay factor	The multiplier between 0 and 1 applied over a specified interval.
doubling time	The input increase required for a model output to multiply by 2.
half-life	The input increase required for a model output to multiply by $1/2$.
extrapolation	Using a model beyond the observed or intended input interval.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A quantity is measured as 50, 61, 74.4, and 90.8 at equally spaced times. Decide whether an exponential model is plausible, estimate the factor, and name one additional check needed before using the model for prediction.

Concept 1: Build from an initial value and rate

Core idea

A percent increase of r gives factor $1 + r$; a percent decrease of r gives factor $1 - r$.

- Use $P(t) = P_0 b^t$ when P_0 is the value at $t = 0$ and b is the factor per input unit.
- If the stated factor applies every k units, use $P(t) = P_0 R^{t/k}$.
- Units should appear in the interpretation even though the factor itself is dimensionless.

Worked example - annotate the reasoning

Problem. A culture starts with 1200 cells and grows by 18% each week. Write a model, predict the population after 7 weeks, and find when it first exceeds 5000 cells.

Model reasoning. $P(t) = 1200(1.18)^t$. $P(7) \approx 3822.57$. Solving $1200(1.18)^t > 5000$ gives $t > 8.62$, so the first whole week is 9.

Check your understanding

Independent

A device worth 4800 QAR loses 14% of its value each year. Write the model and find its value after 5 years.

Concept 2: Build from two points or a characteristic time

Core idea

For $P(t) = ab^t$, the ratio of two outputs determines b after accounting for the input gap.

- $b = \left(\frac{P(t_2)}{P(t_1)} \right)^{1/(t_2 - t_1)}$.
- Doubling time d gives $P(t) = P_0 2^{t/d}$.
- Half-life h gives $P(t) = P_0 (1/2)^{t/h}$.

Worked example - annotate the reasoning

Problem. An exponential model satisfies $P(2) = 360$ and $P(6) = 5760$. Determine the model $P(t) = ab^t$.

Model reasoning. $5760/360 = 16$, so $b^4 = 16$ and $b = 2$. Then $360 = a(2)^2$, giving $a = 90$. Therefore $P(t) = 90(2)^t$.

Check your understanding

Independent

A substance begins at 640 mg and has half-life 9 hours. Find the amount after 24 hours.

Concept 3: Shifted exponential models and limitations

Core idea

A model $P(t) = L + Ab^t$ approaches the limiting value L rather than zero.

- Subtract the limiting value before finding ratios from data.
- Interpret $A = P(0) - L$ as the initial difference from the limit.
- Model fit does not establish causation, and a good short-term fit may fail under long-term extrapolation.

Worked example - annotate the reasoning

Problem. A temperature model approaches 20°C and satisfies $T(0) = 95$ and $T(3) = 58.4$. Find $T(t)$.

Model reasoning. Let $T(t) = 20 + Ab^t$. From $T(0) = 95$, $A = 75$. Then $58.4 - 20 = 38.4 = 75b^3$, so $b^3 = 0.512$ and $b = 0.8$. Thus $T(t) = 20 + 75(0.8)^t$.

Check your understanding

Independent

Why is it unreasonable to extend a bacterial growth model indefinitely in a closed container?

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Write a model for an initial amount of 300 that decreases by 9% per day.

Exit ticket 2

No calculator

A model doubles every 5 hours. By what factor does it change in 15 hours?

Exit ticket 3

No calculator

Interpret the parameter L in $P(t) = L + A(0.7)^t$.

Topic 2.6 - Competing Function Model Validation

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

Model selection is an argument supported by structure, not a contest to find the largest correlation number. Strong validation combines a plausible mechanism, a well-scaled residual analysis, comparison with competing models, and an explicit statement of the intended domain.

Learning targets

- Select plausible linear, quadratic, or exponential models from patterns, context, and residual behavior.
- Compute and interpret signed residuals and absolute error measures.
- Validate a model within a domain and articulate consequences of overprediction, underprediction, and extrapolation.

Language and structure

Term	Meaning in this topic
residual	Observed value minus predicted value: $r = y - \hat{y}$.
underprediction	A positive residual; the model prediction is below the observed value.
overprediction	A negative residual; the model prediction is above the observed value.
mean absolute error	The average of the absolute residuals.
systematic pattern	A nonrandom residual trend suggesting that important structure remains unmodeled.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Three models fit the same five data points: a linear model with MAE 2.8, a quadratic model with MAE 1.1, and an exponential model with MAE 1.3. What additional evidence is needed before selecting the quadratic model?

Concept 1: Choose candidates from structural fingerprints

Core idea

Linear data have constant first differences, quadratic data have constant second differences, and exponential data have constant ratios over equal input intervals.

- Approximate patterns suggest candidates but do not prove a unique model.
- Context can eliminate a mathematically good fit that contradicts the mechanism or allowable outputs.
- Use the same input interval when comparing differences or ratios.

Worked example - annotate the reasoning

Problem. The outputs 10, 15, 22, 31, 42 correspond to inputs 0, 1, 2, 3, 4. Identify a plausible exact model.

Model reasoning. First differences are 5, 7, 9, 11 and second differences are constantly 2, so a quadratic model is appropriate. Solving gives $y = x^2 + 4x + 10$.

Check your understanding

Independent

The outputs are 96, 72, 54, 40.5, 30.375 at consecutive inputs. Identify a plausible model.

Concept 2: Compute and read residuals

Core idea

A residual is $r = y - \hat{y}$; its sign describes the direction of model error.

- Positive residual: the model underpredicts.
- Negative residual: the model overpredicts.
- Small residuals scattered around 0 support a model more than residuals with a curved or one-sided pattern.

Worked example - annotate the reasoning

Problem. For observed values 80, 64, 51, 41, 33 at $x = 0, 1, 2, 3, 4$, compare $E(x) = 80(0.8)^x$ with $L(x) = 80 - 12x$ using residuals.

Model reasoning. For E , predictions are 80, 64, 51.2, 40.96, 32.768 and residuals are 0, 0, -0.2, 0.04, 0.232. For L , predictions are 80, 68, 56, 44, 32 and residuals are 0, -4, -5, -3, 1. The exponential model has much smaller and less patterned residuals.

Check your understanding

Independent

A model predicts 52 when the observed value is 47. Find and interpret the residual.

Concept 3: Validate within a domain

Core idea

A model can be useful on one interval and unacceptable outside it.

- Compare multiple error measures and inspect residual plots rather than relying on a single statistic.
- State consequences: underpredicting demand, cost, or risk may be more serious than overpredicting.
- Extrapolation requires the mechanism and constraints to remain stable.

Worked example - annotate the reasoning

Problem. A demand model has MAE 4.1 on weeks 1–12 and MAE 18.7 on weeks 13–16. What conclusion is justified?

Model reasoning. The model is reasonably accurate on the calibration interval but degrades substantially later. It should not be treated as validated for weeks 13–16 without revision or new evidence.

Check your understanding

Independent

Why might a slightly less accurate model be preferred to the smallest-MAE model?

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

If $y = 31$ and $\hat{y} = 35.5$, compute and interpret the residual.

Exit ticket 2

No calculator

What residual pattern suggests that a linear model is missing curvature?

Exit ticket 3

No calculator

Why should a model's validation domain be stated?

Topic 2.7 - Composition of Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

A composition represents a process applied to the output of another process. The inside function acts first. A complete solution tracks the intermediate value, verifies that it belongs to the outer function's domain, and interprets the order in context.

Learning targets

- Evaluate compositions from formulas, tables, graphs, and contexts.
- Determine composition domains and explain why order matters.
- Construct and decompose composite functions while interpreting units and intermediate quantities.

Language and structure

Term	Meaning in this topic
composition	The function $(f \circ g)(x) = f(g(x))$.
inner function	The function evaluated first in a composition.
outer function	The function applied to the inner output.
composition domain	Inputs allowed by the inner function whose outputs are also allowed by the outer function.
decomposition	Expressing a function as a composition of simpler functions.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A sensor converts a temperature t into a voltage $V(t) = 0.04t + 1.2$. A display converts voltage v into a reading $R(v) = 25v - 30$. Which composition represents the displayed reading from temperature, and what does the intermediate value mean?

Concept 1: Evaluate in the correct order**Core idea**

In $(f \circ g)(x)$, compute $g(x)$ first and then apply f .

- Use parentheses when substituting the entire inner expression.
- For tables or graphs, locate the inner output and use it as the input to the outer function.
- $(f \circ g)(x)$ and $(g \circ f)(x)$ are generally different.

Worked example - annotate the reasoning

Problem. Let $f(x) = 2x^2 - 3$ and $g(x) = x - 4$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Model reasoning. $(f \circ g)(x) = 2(x - 4)^2 - 3$. $(g \circ f)(x) = (2x^2 - 3) - 4 = 2x^2 - 7$. They are not equal.

Check your understanding

Independent

If $f(5) = 12$ and $g(2) = 5$, find $(f \circ g)(2)$.

Concept 2: Determine the domain of a composition**Core idea**

An input must be in the domain of the inner function and the inner output must be in the domain of the outer function.

- Square roots require nonnegative radicands.
- Denominators cannot equal zero.
- Logarithms require positive arguments.

Worked example - annotate the reasoning

Problem. Let $f(u) = \sqrt{u - 1}$ and $g(x) = 5 - x^2$. Find $(f \circ g)(x)$ and its domain.

Model reasoning. $(f \circ g)(x) = \sqrt{4 - x^2}$. Require $4 - x^2 \geq 0$, so the domain is $[-2, 2]$.

Check your understanding

Independent

Let $p(u) = 1/(u + 3)$ and $q(x) = 2x - 1$. Find the domain of $p(q(x))$.

Concept 3: Build and interpret a composite process

Core idea

Composition is useful when one output becomes the input to another rule.

- Name the intermediate quantity and its units.
- Order reflects the actual sequence of operations.
- Decomposition is not unique, but a useful decomposition should reveal the process.

Worked example - annotate the reasoning

Problem. A store applies a 15% discount and then a 5% service charge. Let $D(x) = 0.85x$ and $S(x) = 1.05x$. Write the final cost and compare it with applying the service charge first.

Model reasoning. $S(D(x)) = 1.05(0.85x) = 0.8925x$. Because multiplication commutes, $D(S(x))$ is also $0.8925x$ in this special case, though composition is not generally commutative.

Check your understanding

Independent

Decompose $H(x) = \sqrt{3x^2 + 7}$ into two simple functions.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

For $f(x) = x^2 + 1$ and $g(x) = 3x$, find $(f \circ g)(2)$.

Exit ticket 2

No calculator

Find the domain of $\sqrt{7 - 2x}$.

Exit ticket 3

No calculator

Why can $f \circ g$ exist at an input where $g \circ f$ does not?

Topic 2.8 - Inverse Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

An inverse function reverses a one-to-one relationship. The domain of the original becomes the range of the inverse, and the range of the original becomes the inverse domain. Restrictions are part of the definition, not optional details.

Learning targets

- Determine whether a function is invertible on a stated domain.
- Construct and verify inverse functions algebraically, numerically, and graphically.
- Interpret inverse inputs, outputs, domains, ranges, and restrictions in context.

Language and structure

Term	Meaning in this topic
one-to-one	A function in which no two allowed inputs produce the same output.
inverse function	A function that reverses the input-output pairing of another function.
horizontal line test	A graphical test for whether a function is one-to-one.
domain restriction	A smaller domain chosen so a function becomes one-to-one.
inverse verification	Showing $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$ on the appropriate domains.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A machine maps an input setting to an output reading. Settings 2 and 8 both produce reading 11. Can the machine's rule have an inverse function on a domain containing both settings? Explain the practical meaning.

Concept 1: Invertibility and restrictions

Core idea

A function has an inverse function exactly when it is one-to-one on its stated domain.

- Use the horizontal line test on a graph.
- A quadratic is not one-to-one on all real numbers but can be restricted to one side of its vertex.
- State the restricted domain before writing the inverse.

Worked example - annotate the reasoning

Problem. Determine whether $f(x) = (x - 2)^2 + 1$ is invertible on $\mathbb{R}$ and on $[2, \infty)$.

Model reasoning. On all real numbers it is not one-to-one because symmetric inputs give the same output. On $[2, \infty)$ it is increasing and one-to-one, so an inverse exists.

Check your understanding**Independent**

Give a domain restriction that makes $g(x) = -(x + 1)^2 + 5$ invertible.

Concept 2: Construct and verify an inverse**Core idea**

Write $y = f(x)$, interchange x and y , solve for y , and carry the domains and ranges with the result.

- Linear and rational functions often invert algebraically without a branch choice.
- Even roots introduce a plus-or-minus choice that is resolved by the original domain.
- Verify by composition and by checking that points (a, b) become (b, a) .

Worked example - annotate the reasoning

Problem. Find the inverse of $f(x) = \frac{2x + 5}{x - 3}$.

Model reasoning. Let $y = (2x + 5)/(x - 3)$. Then $yx - 3y = 2x + 5$, so $x(y - 2) = 3y + 5$ and $x = (3y + 5)/(y - 2)$. Thus $f^{-1}(x) = \frac{3x + 5}{x - 2}$, with inverse domain $x \neq 2$.

Check your understanding

Independent

Find the inverse of $h(x) = (x - 2)^2 + 1$ on $x \geq 2$.

Concept 3: Graph and context

Core idea

The graphs of f and f^{-1} are reflections across the line $y = x$.

- x - and y -intercepts swap roles when they are defined.
- In context, the inverse answers the reverse question: output-to-input rather than input-to-output.
- Units swap: if f maps hours to distance, f^{-1} maps distance to hours.

Worked example - annotate the reasoning

Problem. A conversion rule maps temperature C in degrees Celsius to $F(C) = 1.8C + 32$ degrees Fahrenheit. Find and interpret the inverse.

Model reasoning. $F^{-1}(x) = (x - 32)/1.8$. It takes a Fahrenheit temperature as input and returns the corresponding Celsius temperature.

Check your understanding

Independent

If $(4, -7)$ lies on f , what point lies on f^{-1} ?

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1**No calculator**

Find the inverse of $f(x) = 5x - 9$.

Exit ticket 2**No calculator**

State the inverse domain of $g(x) = \sqrt{x+4} - 2$.

Exit ticket 3**No calculator**

What graph transformation connects f and f^{-1} ?

Topic 2.9 - Logarithmic Expressions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

A logarithm answers an exponent question. Every valid logarithm has a positive argument, a positive base different from 1, and a real output. Exact evaluation relies on recognizing powers; approximation uses change of base or calculator functions while preserving domain restrictions.

Learning targets

- Evaluate logarithmic expressions by translating between logarithmic and exponential forms.
- Approximate common, natural, and arbitrary-base logarithms with technology.
- Interpret logarithms as exponents in mathematical, scientific, and contextual scales.

Language and structure

Term	Meaning in this topic
logarithm	The exponent to which a base must be raised to produce a given positive number.
common logarithm	A logarithm with base 10, written $\log x$.
natural logarithm	A logarithm with base e , written $\ln x$.
argument	The positive expression inside a logarithm.
change of base	The identity $\log_b x = \frac{\ln x}{\ln b} = \frac{\log x}{\log b}$.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Without a calculator, place $\log_2 20$ between two consecutive integers and explain the reasoning in exponential language.

Concept 1: Logarithmic and exponential forms

Core idea

$\log_b a = c$ if and only if $b^c = a$, where $b > 0$, $b \neq 1$, and $a > 0$.

- The base stays the same when moving between forms.
- The logarithm output is an exponent.
- Use the exponential form to solve simple equations involving an unknown argument, exponent, or base.

Worked example - annotate the reasoning

Problem. Rewrite $\log_3 81 = 4$ in exponential form and evaluate $\log_{1/2} 8$.

Model reasoning. $\log_3 81 = 4$ means $3^4 = 81$. For $\log_{1/2} 8 = c$, solve $(1/2)^c = 8 = 2^3$, so $2^{-c} = 2^3$ and $c = -3$.

Check your understanding**Independent**

Solve $\log_b 125 = 3$ for the positive base b .

Concept 2: Exact values and restrictions**Core idea**

Exact logarithms come from known powers and roots; the argument must remain positive.

- $\log_b 1 = 0$ because $b^0 = 1$.
- $\log_b b = 1$ because $b^1 = b$.
- Fractional arguments can produce negative logarithms when the base is greater than 1.

Worked example - annotate the reasoning

Problem. Evaluate $\log_4(1/32)$ exactly.

Model reasoning. Since $4 = 2^2$ and $1/32 = 2^{-5}$, solve $2^{2c} = 2^{-5}$, giving $c = -5/2$.

Check your understanding

Independent

State the domain of $\log_5(2x - 7)$.

Concept 3: Approximation and scale meaning

Core idea

Use change of base when the calculator does not have the desired base.

- Round only at the end of a calculation.
- A difference of 1 on a base-10 logarithmic scale corresponds to a factor of 10 in the underlying quantity.
- Interpret the result as an exponent or scale level, not as a raw amount.

Worked example - annotate the reasoning

Problem. Approximate $\log_7 250$ to three decimal places.

Model reasoning. $\log_7 250 = \ln 250 / \ln 7 \approx 2.838$.

Check your understanding

Independent

The pH model is $\text{pH} = -\log[H^+]$. Compare hydrogen-ion concentrations for pH 4 and pH 6.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1**No calculator**Evaluate $\log_9 3$.

Exit ticket 2**No calculator**Rewrite $5^{-2} = 1/25$ in logarithmic form.

Exit ticket 3**No calculator**State the domain of $\ln(4 - x)$.

Topic 2.10 - Inverses of Exponential Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

Logarithmic functions arise by reversing exponential functions. The horizontal asymptote of the exponential becomes a vertical asymptote of the logarithm, and the exponential range becomes the logarithm domain. A transformed inverse must preserve every shift, scale, and restriction.

Learning targets

- Explain logarithms as inverse functions of exponentials.
- Connect inverse pairs through compositions, points, graphs, domains, ranges, and asymptotes.
- Construct inverses of basic and transformed exponential functions.

Language and structure

Term	Meaning in this topic
inverse pair	Two functions whose compositions return the original input on appropriate domains.
exponential-log identity	$\log_b(b^x) = x$ and $b^{\log_b x} = x$ for $x > 0$.
vertical asymptote	A vertical line approached by a logarithmic graph near the edge of its domain.
domain-range swap	The domain of one inverse partner equals the range of the other.
reflection	The graphs of inverse functions are reflections across $y = x$.

Launch: make a claim before calculating

Launch investigation

Reasoning first

The function $f(x) = 2^x$ maps 3 to 8. What must its inverse do with 8? Use this fact to explain the meaning of $\log_2 8$.

Concept 1: Exponential and logarithmic identities

Core idea

For $b > 0$, $b \neq 1$, the inverse of $f(x) = b^x$ is $f^{-1}(x) = \log_b x$.

- $\log_b(b^x) = x$ for every real x .
- $b^{\log_b x} = x$ only for $x > 0$.
- These identities are composition statements, not product rules.

Worked example - annotate the reasoning

Problem. Simplify $\log_5(5^{2x-3})$ and $5^{\log_5(4x+1)}$, stating any restriction.

Model reasoning. The first is $2x - 3$ for all real x . The second is $4x + 1$ provided $4x + 1 > 0$, so $x > -1/4$.

Check your understanding

Independent

Simplify $\ln(e^{x^2+2})$ and $e^{\ln(7-x)}$.

Concept 2: Points, graphs, and asymptotes

Core idea

If (a, b) lies on an exponential function, then (b, a) lies on its logarithmic inverse.

- The exponential domain $\mathbb{R}$ becomes the logarithmic range $\mathbb{R}$.
- The exponential range (k, ∞) or $(-\infty, k)$ becomes the inverse domain.
- The horizontal asymptote $y = k$ reflects to the vertical asymptote $x = k$.

Worked example - annotate the reasoning

Problem. For $f(x) = 2^x + 3$, state the inverse, its domain and range, and its vertical asymptote.

Model reasoning. $f^{-1}(x) = \log_2(x - 3)$. Domain $(3, \infty)$, range $\mathbb{R}$, and vertical asymptote $x = 3$.

Check your understanding

Independent

If $(4, 85)$ lies on $g(x) = 5(2)^x + 5$, identify the corresponding inverse point.

Concept 3: Inverse of a transformed exponential

Core idea

Isolate the exponential expression before applying a logarithm.

- For $f(x) = ab^{x-h} + k$, solve $\frac{y-k}{a} = b^{x-h}$.
- The sign of a determines which side of k forms the range and inverse domain.
- Check the inverse by composition or by reversing a known point.

Worked example - annotate the reasoning

Problem. Find the inverse of $f(x) = 3(2)^{x-1} + 4$.

Model reasoning. $y - 4 = 3(2)^{x-1}$, so $(y - 4)/3 = 2^{x-1}$ and $x = 1 + \log_2((y - 4)/3)$. Thus $f^{-1}(x) = 1 + \log_2((x - 4)/3)$, domain $x > 4$.

Check your understanding

Independent

Find the inverse of $g(x) = 5 - 2(3)^x$.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1**No calculator**

Find the inverse of $f(x) = 4^x - 6$.

Exit ticket 2**No calculator**

State the vertical asymptote of the inverse of $g(x) = 7(0.5)^x + 2$.

Exit ticket 3**No calculator**

Simplify $3^{\log_3(x-1)}$ and state the domain.

Topic 2.11 - Logarithmic Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

A logarithmic function compresses multiplicative input change into additive output change. Its vertical asymptote marks the boundary of the domain, and transformations must be interpreted through the argument as well as the outside coefficient.

Learning targets

- Describe logarithmic domain, range, asymptotes, monotonicity, concavity, and end behavior.
- Connect equal multiplicative changes in input to equal additive changes in output.
- Analyze and construct transformed logarithmic functions with explicit restrictions.

Language and structure

Term	Meaning in this topic
vertical asymptote	A vertical line approached as the logarithm argument tends to 0^+ .
multiplicative interval	An input change by a constant factor rather than a constant difference.
anchor point	For $a \log_b(x - h) + k$, the point $(h + 1, k)$ because $\log_b 1 = 0$.
monotonicity	Increasing or decreasing behavior determined by the base and any reflection.
concavity	For $\log_b x$, concave down when $b > 1$ and concave up when $0 < b < 1$.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A function changes by $+3$ whenever its positive input is multiplied by 10. What family is suggested, and what information is still needed to determine a unique formula?

Concept 1: The parent logarithmic family

Core idea

For $f(x) = \log_b x$, the domain is $(0, \infty)$, the range is $\mathbb{R}$, and the vertical asymptote is $x = 0$.

- If $b > 1$, the function is increasing and concave down.
- If $0 < b < 1$, the function is decreasing and concave up.
- The graph passes through $(1, 0)$ and $(b, 1)$.

Worked example - annotate the reasoning

Problem. Analyze $f(x) = \log_3 x$: state domain, range, intercept, asymptote, monotonicity, concavity, and end behavior.

Model reasoning. Domain $(0, \infty)$, range $\mathbb{R}$, x -intercept $(1, 0)$, no y -intercept, asymptote $x = 0$, increasing and concave down. As $x \rightarrow 0^+$, $f(x) \rightarrow -\infty$; as $x \rightarrow \infty$, $f(x) \rightarrow \infty$.

Check your understanding**Independent**

Analyze $g(x) = \log_{1/2} x$.

Concept 2: Transformations and domain**Core idea**

For $f(x) = a \log_b(x - h) + k$, the domain is $x > h$, the asymptote is $x = h$, and the anchor point is $(h + 1, k)$.

- The sign of a can reverse monotonicity and concavity.
- Horizontal behavior comes from setting the argument equal to familiar powers of the base.
- Never state the domain from the outside shift; solve the argument inequality.

Worked example - annotate the reasoning

Problem. Analyze $f(x) = -3 \log_4(x - 2) + 5$.

Model reasoning. Domain $(2, \infty)$; range $\mathbb{R}$; asymptote $x = 2$; anchor point $(3, 5)$. Since $\log_4$ is increasing and concave down, multiplication by -3 makes f decreasing and concave up.

Check your understanding

Independent

For $p(x) = 2 \ln(5 - x) - 1$, state the domain and asymptote.

Concept 3: Multiplicative input change

Core idea

For $f(x) = a \log_b x + k$, multiplying the input by b adds a to the output.

- $f(bx) - f(x) = a$.
- More generally, multiplying input by b^m changes output by am .
- This property explains why logarithmic scales convert ratios into differences.

Worked example - annotate the reasoning

Problem. For $f(x) = 4 \log_2 x - 3$, compare $f(8x)$ with $f(x)$.

Model reasoning. $8 = 2^3$, so $f(8x) - f(x) = 4 \cdot 3 = 12$. Thus $f(8x) = f(x) + 12$.

Check your understanding

Independent

A logarithmic model increases by 6 whenever the input is multiplied by 5. Write one possible model.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

State the domain and asymptote of $f(x) = \log_7(x + 4) - 2$.

Exit ticket 2

No calculator

Find the anchor point of $g(x) = 5 \ln(x - 3) + 1$.

Exit ticket 3

No calculator

For $h(x) = 2 \log_3 x$, how does $h(9x)$ compare with $h(x)$?

Topic 2.12 - Logarithmic Function Manipulation

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

Logarithmic properties transform multiplication into addition, division into subtraction, and powers into coefficients. These identities are valid only when every logarithm involved has a positive argument. A correct rewrite includes the same domain as the original expression.

Learning targets

- Expand and condense logarithmic expressions using product, quotient, and power properties.
- Use change of base and natural logarithms to evaluate and compare logarithms.
- Explain algebraic and graphical equivalences while preserving domain restrictions.

Language and structure

Term	Meaning in this topic
product property	$\log_b(MN) = \log_b M + \log_b N$ for $M, N > 0$.
quotient property	$\log_b(M/N) = \log_b M - \log_b N$ for $M, N > 0$.
power property	$\log_b(M^p) = p \log_b M$ when the expressions are defined.
condense	Combine several logarithms into one logarithm.
domain equivalence	A rewrite must preserve the set of allowed inputs, not only the algebraic appearance.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A student claims $\log(x + 4) = \log x + \log 4$. Test the claim at one convenient input and then state the correct property that may have caused the confusion.

Concept 1: Expand with structure

Core idea

Factor products, quotients, and powers before applying logarithmic properties.

- Keep numerical constants that are powers of the base exact when possible.
- State assumptions such as $x > 0$ when a power property introduces $\log x$.
- Do not split logarithms across addition or subtraction inside the argument.

Worked example - annotate the reasoning

Problem. Expand $\log_3 \left(\frac{9x^2}{\sqrt{y}} \right)$ assuming $x > 0$ and $y > 0$.

Model reasoning. $\log_3 9 + 2 \log_3 x - \frac{1}{2} \log_3 y = 2 + 2 \log_3 x - \frac{1}{2} \log_3 y$.

Check your understanding

Independent

Expand $\ln \left(\frac{(x-1)^4 \sqrt{x+2}}{5} \right)$ on its domain.

Concept 2: Condense and preserve the domain

Core idea

Coefficients become exponents; sums become products and differences become quotients.

- Start by moving coefficients into powers.
- Combine only after identifying the individual logarithm restrictions.
- A condensed expression can appear defined at extra inputs; retain the original domain explicitly.

Worked example - annotate the reasoning

Problem. Condense $2 \ln(x-1) - \frac{1}{2} \ln(x+2) + \ln 5$.

Model reasoning. $\ln \left(\frac{5(x-1)^2}{\sqrt{x+2}} \right)$, with original domain $x > 1$.

Check your understanding

Independent

Condense $\log_2 a + 3 \log_2 b - \log_2 c$ for positive a, b, c .

Concept 3: Change of base and graph equivalence

Core idea

$\log_b x = \frac{\ln x}{\ln b}$ allows any valid base to be evaluated or compared.

- Changing the base does not change the function value.
- $\log_b(cx) = \log_b x + \log_b c$ for $c > 0$ and $x > 0$.
- $\log_b(x^p) = p \log_b x$ on the shared domain; this can describe vertical scaling.

Worked example - annotate the reasoning

Problem. Rewrite $\log_5 x$ using natural logarithms and explain why $\log_{25} x = \frac{1}{2} \log_5 x$.

Model reasoning. $\log_5 x = \ln x / \ln 5$. Since $\ln 25 = 2 \ln 5$, $\log_{25} x = \ln x / (2 \ln 5) = \frac{1}{2} \log_5 x$.

Check your understanding

Independent

Compare the graphs of $f(x) = \log_2(8x)$ and $g(x) = \log_2 x$.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Expand $\log_2(8x^3)$ for $x > 0$.

Exit ticket 2

No calculator

Condense $\ln a - \ln b + 2 \ln c$ for positive variables.

Exit ticket 3

No calculator

Rewrite $\log_7 20$ using natural logarithms.

Topic 2.13 - Exponential and Logarithmic Equations and Inequalities

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Why this topic matters

Exponential and logarithmic equations are solved by exposing the inverse structure. Exact solutions often use common bases or logarithms; approximate solutions use technology only after a correct algebraic setup. Logarithmic equations require a final domain check because algebraic manipulation can create invalid candidates.

Learning targets

- Solve exponential equations and inequalities exactly or approximately.
- Solve logarithmic equations and inequalities while enforcing domain restrictions.
- Interpret solutions and thresholds in transformed functions and contextual models.

Language and structure

Term	Meaning in this topic
common-base method	Rewrite both sides with the same exponential base and equate exponents.
logarithmic method	Apply a logarithm to isolate an exponent.
extraneous solution	A candidate produced algebraically that violates an original equation or domain.
monotonic inequality	An inequality solved using whether an exponential or logarithmic function is increasing or decreasing.
threshold	An input at which a modeled quantity first reaches or passes a specified level.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Compare the strategies for solving $2^{x+1} = 7$ and $2^{x+1} = 16$. Which solution can be exact without logarithms, and why?

Concept 1: Exponential equations

Core idea

Use common bases when possible; otherwise isolate the exponential and apply a logarithm.

- If $b^u = b^v$ with $b > 0$, $b \neq 1$, then $u = v$.
- $b^x = c$ gives $x = \log_b c = \ln c / \ln b$.
- Substitution such as $u = b^x$ can turn some equations into quadratics.

Worked example - annotate the reasoning

Problem. Solve $3^{2x} - 10(3^x) + 9 = 0$.

Model reasoning. Let $u = 3^x > 0$. Then $u^2 - 10u + 9 = (u - 1)(u - 9) = 0$, so $u = 1$ or $u = 9$. Thus $x = 0$ or $x = 2$.

Check your understanding

Independent

Solve $5^{x-1} = 17$ exactly and approximately.

Concept 2: Logarithmic equations and domain checks

Core idea

Condense logs when helpful, convert to exponential form, solve, and test every candidate in the original domain.

- Each logarithm argument must be positive.
- Squaring or multiplying expressions can introduce candidates that do not satisfy the original equation.
- Check exact candidates before rounding.

Worked example - annotate the reasoning

Problem. Solve $\log_2(x - 1) + \log_2(x - 5) = 3$.

Model reasoning. Domain $x > 5$. Condense: $\log_2((x - 1)(x - 5)) = 3$, so $(x - 1)(x - 5) = 8$. This gives $x = 3 \pm 2\sqrt{3}$. Only $x = 3 + 2\sqrt{3}$ satisfies $x > 5$.

Check your understanding

Independent

Solve $\ln(x+2) - \ln(x-1) = \ln 3$.

Concept 3: Inequalities and thresholds

Core idea

Use monotonicity: bases greater than 1 preserve inequality direction; bases between 0 and 1 reverse it.

- Logarithms with base greater than 1 are increasing; bases between 0 and 1 are decreasing.
- Intersect the algebraic solution with the original domain.
- In discrete contexts, convert a real threshold time to the first allowed whole-number input.

Worked example - annotate the reasoning

Problem. Solve $(1/2)^{x-2} \geq 8$.

Model reasoning. Write $8 = (1/2)^{-3}$. Since base $1/2$ is decreasing, $x - 2 \leq -3$, so $x \leq -1$.

Check your understanding

Independent

A balance follows $B(t) = 2500(1.07)^t$. Find the first whole year when $B(t) > 4000$.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1**No calculator**Solve $4^{x-1} = 64$.

Exit ticket 2**No calculator**Solve $\log_3(x + 1) = 2$.

Exit ticket 3**No calculator**Solve $\log_2(x - 4) > 1$.

Topic 2.14 - Logarithmic Function Context and Data Modeling

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Why this topic matters

Logarithmic models are appropriate when equal multiplicative changes in input produce approximately equal additive changes in output. They often describe diminishing returns or scale measurements, but their restricted domains and slow long-run growth must be interpreted in context.

Learning targets

- Construct logarithmic models from proportional input change, two points, transformations, or regression output.
- Use logarithmic models to predict outputs and solve for inputs.
- Interpret domains, units, scale meaning, diminishing change, and model limitations.

Language and structure

Term	Meaning in this topic
reference input	A convenient positive input at which the logarithm is 0, often $x = 1$ or $x = h + 1$.
diminishing returns	Output continues to increase while gains become smaller for equal additive input changes.
multiplicative predictor	An input whose meaningful comparisons are often ratios rather than differences.
logarithmic regression model domain	A fitted model such as $y = a + b \ln x$ for positive inputs. The positive or shifted-positive input interval on which the context and formula are meaningful.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A performance measure rises by about 6 points when practice time increases from 1 to 2 hours, from 2 to 4 hours, and from 4 to 8 hours. What model family is suggested, and why is a linear model less natural?

Concept 1: Build from multiplicative change

Core idea

If multiplying x by b changes output by a constant a , a model $f(x) = a \log_b x + k$ is natural.

- $f(bx) - f(x) = a$.
- The value at $x = 1$ is k because $\log_b 1 = 0$.
- Changing the base changes the coefficient but can represent the same function.

Worked example - annotate the reasoning

Problem. A score is 5 at input 1 and increases by 4 whenever the input doubles. Construct a model.

Model reasoning. $S(x) = 4 \log_2 x + 5$ for $x > 0$.

Check your understanding

Independent

A response increases by 9 whenever the input is multiplied by 10, and the response is 22 at input 1. Write a model.

Concept 2: Build from two points and transformations

Core idea

For $f(x) = a \log_b(x-h) + k$, use the asymptote $x = h$ and points whose shifted inputs are convenient powers of b .

- The anchor point is $(h+1, k)$.
- Subtract k before comparing output changes.

- All modeled inputs must satisfy $x > h$.

Worked example - annotate the reasoning

Problem. Construct $f(x) = a \log_2 x + k$ through $(1, 5)$ and $(8, 17)$.

Model reasoning. At $x = 1$, $k = 5$. At $x = 8$, $\log_2 8 = 3$, so $3a + 5 = 17$ and $a = 4$. Thus $f(x) = 4 \log_2 x + 5$.

Check your understanding

Independent

A model has vertical asymptote $x = 3$, passes through $(4, 7)$, and increases by 6 when $x - 3$ is multiplied by 5. Find the model.

Concept 3: Prediction, inverse use, and limitations**Core idea**

To solve for an input, isolate the logarithm and convert to exponential form.

- Predictions near the domain boundary can be unstable because the logarithm changes rapidly.
- Long-run logarithmic growth is slow but unbounded; physical limits may still invalidate extrapolation.
- A good fit does not imply that a logarithmic mechanism causes the relationship.

Worked example - annotate the reasoning

Problem. For $S(x) = 30 + 8 \ln x$, find the input that gives output 50.

Model reasoning. $8 \ln x = 20$, so $\ln x = 2.5$ and $x = e^{2.5} \approx 12.182$.

Check your understanding

Independent

Why is $S(x) = 30 + 8 \ln x$ not meaningful at $x = 0$?

Synthesis and exit ticket**Before you leave**

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Construct a model with $f(1) = 12$ that increases by 5 when input is multiplied by 3.

Exit ticket 2

No calculator

Find the domain of $g(x) = 7 + 2 \ln(x - 4)$.

Exit ticket 3

No calculator

Solve $10 + 3 \log_2 x = 25$.

Topic 2.15 - Semi-log Plots

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

A semi-log plot makes exponential data linear by plotting the input against the logarithm of a positive output. Equal vertical distances represent equal ratios, not equal differences. The line's intercept determines the exponential coefficient, and its slope determines the growth or decay factor.

Learning targets

- Interpret semi-log plots with a logarithmically scaled output axis.
- Use linearity in x versus $\log y$ to identify and construct exponential models.
- Explain how slope, intercept, base, shifts, and limitations affect linearization.

Language and structure

Term	Meaning in this topic
semi-log plot	A graph with one logarithmic axis and one ordinary linear axis.
linearization	A transformation that turns a nonlinear relationship into a line.
log-output coordinate	The plotted value $z = \log y$ or $z = \ln y$ rather than y itself.
slope-factor link	If $\log_b y = mx + c$, then $y = b^c(b^m)^x$.
baseline shift	A limiting value L that must often be subtracted before taking a logarithm.

Launch: make a claim before calculating

Launch investigation

Reasoning first

On a base-10 logarithmic vertical axis, the labels 1, 10, 100, and 1000 are equally spaced. Why does this not mean the output changes by an equal amount between consecutive labels?

Concept 1: Why exponentials become lines

Core idea

If $y = ab^x$ with $a > 0$, then $\log y = \log a + x \log b$.

- On a plot of x versus $\log y$, the intercept is $\log a$.
- The slope is $\log b$ for the same log base used on the axis.
- A positive slope means growth; a negative slope means decay.

Worked example - annotate the reasoning

Problem. Linearize $y = 5(2)^x$ using common logarithms and identify the line's slope and intercept.

Model reasoning. Let $z = \log y$. Then $z = \log 5 + x \log 2$. The intercept is $\log 5 \approx 0.6990$ and the slope is $\log 2 \approx 0.3010$.

Check your understanding**Independent**

If $\ln y = 1.2 - 0.3x$, write y as an exponential function.

Concept 2: Read slope and intercept from a semi-log line**Core idea**

If $z = \log_{10} y = mx + c$, then $a = 10^c$ and the one-unit factor is 10^m .

- A change of Δx multiplies y by $10^{m\Delta x}$.
- Vertical intervals labeled 1, 10, 100, 1000 are equally spaced on a base-10 log axis.
- Use actual log coordinates rather than visual ordinary-axis distances.

Worked example - annotate the reasoning

Problem. A semi-log line satisfies $\log y = 0.7 + 0.15x$. Find the exponential model and interpret the one-unit factor.

Model reasoning. $y = 10^{0.7}(10^{0.15})^x \approx 5.012(1.413)^x$. Each one-unit increase in x multiplies y by about 1.413.

Check your understanding

Independent

On a base-10 semi-log plot, the line rises 0.6 log units when x increases by 4. Find the factor per unit.

Concept 3: Shifts, diagnostics, and limitations

Core idea

A shifted exponential $y = L + ab^x$ is linearized by plotting $\log(y - L)$, not usually by plotting $\log y$.

- All transformed outputs must be positive.
- Curvature on a semi-log plot suggests that a single unshifted exponential is not adequate.
- Linearity over a short interval does not guarantee reliable extrapolation.

Worked example - annotate the reasoning

Problem. Explain how to linearize $T(t) = 20 + 70(0.8)^t$.

Model reasoning. Subtract the baseline: $T - 20 = 70(0.8)^t$. Then $\ln(T - 20) = \ln 70 + t \ln 0.8$, which is linear in t .

Check your understanding

Independent

A semi-log plot bends downward rather than forming a line. Give two possible explanations.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1**No calculator**

If $\log y = 0.4x + 1$, write y in exponential form.

Exit ticket 2**No calculator**

What is the semi-log slope for $y = 12(0.7)^x$ using natural logarithms?

Exit ticket 3**No calculator**

Why must $y > 0$ before plotting $\log y$?
